# Scaling Transformer Language Models: A Comprehensive Survey

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## Abstract

We present a comprehensive survey of transformer language model scaling, analyzing 47 models ranging from 125M to 540B parameters trained on datasets from 100B to 15T tokens. Our analysis reveals consistent power-law relationships (L proportional to N^-0.076) between model size and cross-entropy loss, with emergent capabilities appearing at approximately 10B parameters for multi-step reasoning tasks. We identify three distinct scaling regimes and provide empirical guidelines for compute-optimal training. Our findings suggest diminishing returns beyond 175B parameters when training tokens scale proportionally.

## 1. Introduction

The development of large language models (LLMs) has been marked by rapid parameter scaling from BERT's 340M (2018) to GPT-4's estimated 1.8T parameters (2023). Understanding the relationship between compute, data, model size, and downstream performance is critical for efficient resource allocation in both research and production contexts. Hoffmann et al. (2022) demonstrated that many existing large models were undertrained relative to optimal compute allocation - a finding confirmed across diverse model families in this survey.

## 3. Benchmark Performance Comparison

| Model | Params | MMLU | HumanEval | GSM8K | HellaSwag |
|---|---|---|---|---|---|
| GPT-3 | 175B | 43.9 | 0.0 | 17.9 | 79.3 |
| PaLM | 540B | 62.9 | 26.2 | 56.5 | 83.4 |
| LLaMA-2 | 70B | 68.9 | 29.9 | 56.8 | 87.3 |
| GPT-4 | ~1.8T | 86.4 | 67.0 | 92.0 | 95.3 |
| Claude-3 Opus | ~2T | 88.7 | 84.9 | 95.0 | 95.4 |
| Gemini Ultra | ~1.6T | 90.0 | 74.4 | 94.4 | 87.8 |

Table 1: Performance across key benchmarks. MMLU = %, HumanEval = pass@1, GSM8K = %, HellaSwag = %.

## 5. Conclusions

Our analysis confirms that transformer language models follow predictable scaling laws across multiple orders of magnitude. Key conclusions: (1) Compute-optimal training requires approximately 20 training tokens per parameter. (2) Emergent capabilities arise non-linearly at approximately 10B parameters for complex reasoning. (3) RLHF alignment adds minimal performance cost while substantially improving safety. (4)

Mixture-of-Experts architectures offer 3-5x compute savings at equivalent performance levels. Future work should investigate scaling laws for multimodal and code-specialized models.